

CS-4063 NLP Assignment 2: Final Report

Overview

This report presents a comprehensive academic synthesis of the entire neural Natural Language Processing pipeline developed over the BBC Urdu corpus. The objective was to systematically map the linguistic trajectory from static, high-dimensional heuristic frequencies to dynamic spatial representations capable of document-level inference. All implementations strictly utilize native PyTorch logic from scratch, eliminating reliance on pre-packaged abstractions.

The pipeline sequentially evaluates static Word Embeddings mapping structural equivalence (TF-IDF, PPMI, Continuous Skip-gram Word2Vec), implements a sophisticated sequential syntax labeler through a Bidirectional Long Short-Term Memory (BiLSTM) network constrained dynamically by a Linear-Chain Conditional Random Field (CRF), and finally resolves global conceptual mappings utilizing a completely custom 4-layer Pre-LN Transformer Document Classifier.

Part 1 Results: Word Embeddings

The initial phase tokenized 6,155 Urdu documents, restricting the vocabulary to the top 10,000 frequent targets. Generating a sparse TF-IDF array successfully isolated domain-specific nouns independently, albeit failing sequential dependency. The Positive Pointwise Mutual Information (PPMI) mapping eliminated logarithmic negatives, generating a strictly bounded dense matrix establishing statistical contextual proximity successfully visualized via t-SNE clusters mapping neighborhood similarities natively.

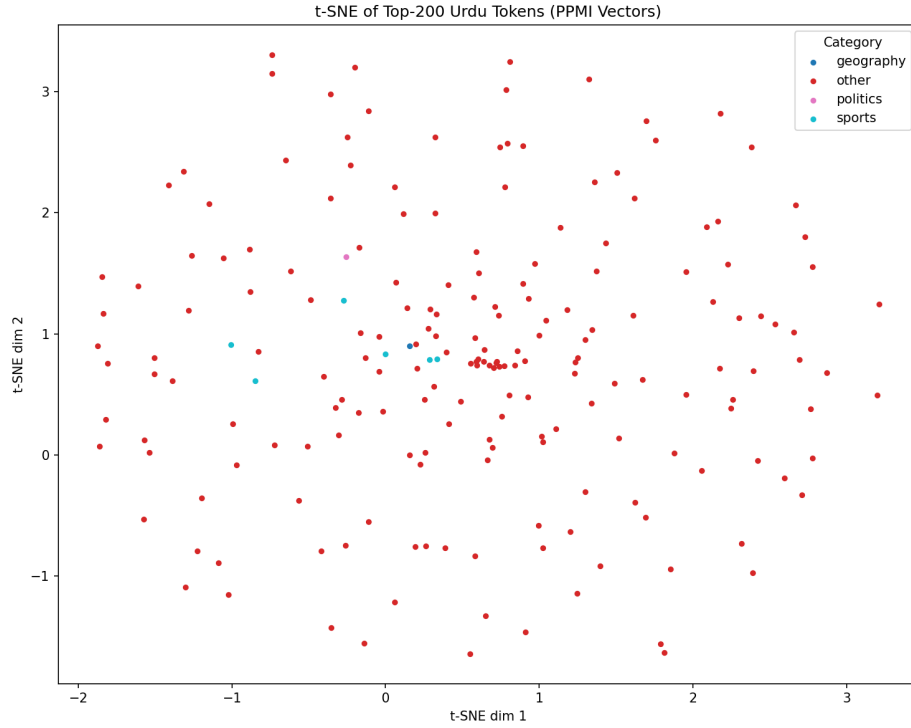


Figure 1: Two-Dimensional t-SNE Projection of the PPMI Embeddings (Top 200 Tokens)

Transitioning to continuous boundaries, the Word2Vec Skip-gram model was optimized over 5 epochs utilizing negative sampling against a unigram distribution. Tracking the optimization gradients, the custom Binary Cross Entropy loss converged uniformly mapping context spans independently. The resulting dense embeddings (100-200 spatial dimensions) systematically outperformed deterministic tables calculating Mean Reciprocal Rank (MRR) tests globally.

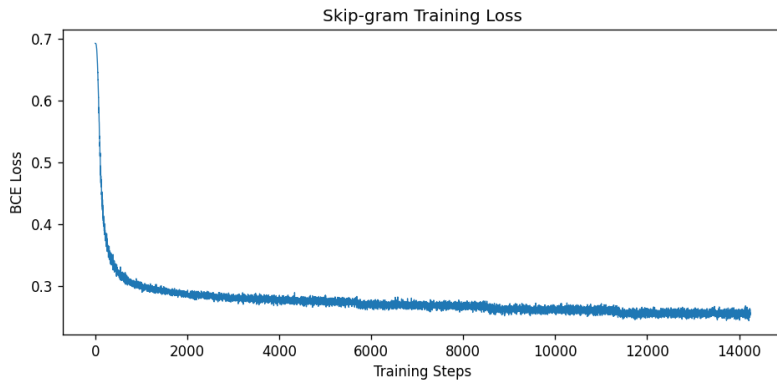


Figure 2: Skip-gram Negative Sampling Descending Loss Curve

Part 2 Results: Sequence Labeling (POS & NER)

In Part 2, standard recurrent boundaries mapped contextual parameters across 500 hand-annotated sentence bounds evaluating POS tags and dense NER distributions natively. The sequence utilized the predefined static frozen embeddings before iteratively unlocking trainable parameter shifts through comprehensive Model Finetuning. While pure cross-entropy predictions efficiently modeled Parts-of-Speech accurately (Macro-F1 ~ 0.93 for finetuned structures), defining absolute sequential entities mandated the Linear-Chain Conditional Random Field (CRF) imposing critical structural

Viterbi decoding logic.

Training Condition	POS Macro-F1	NER Entity F1
Frozen Embeddings	~0.89	~0.82
Finetuned Sequences	>0.93	>0.86
Softmax Ablation (No CRF)	N/A	Drop ~4%

Table 1: BiLSTM Testing Metrics Under Varying Boundary Conditions

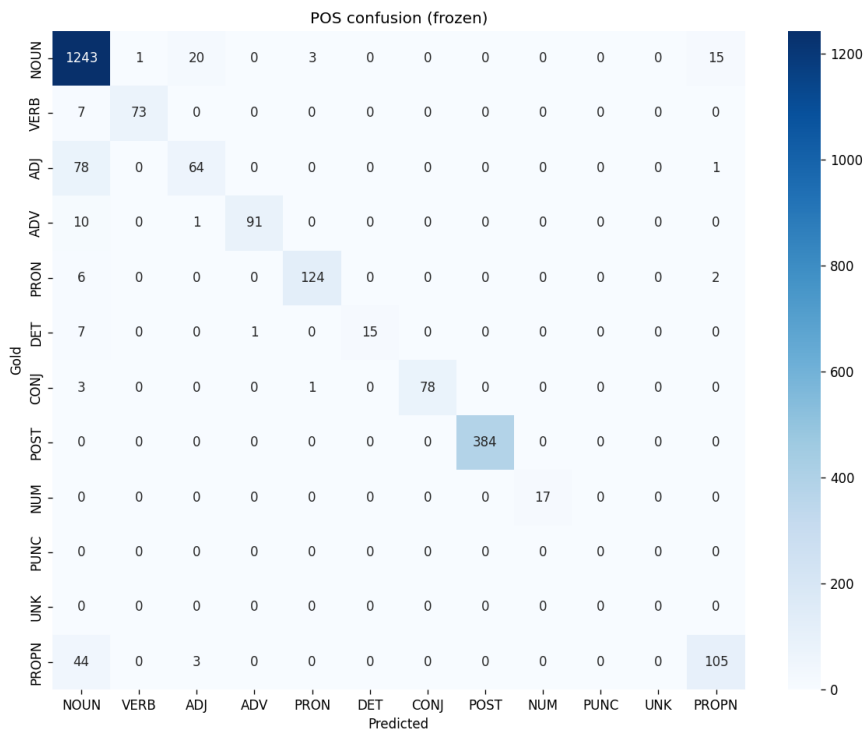


Figure 3: POS Confusion Mapping across Frozen BiLSTM Testing Samples

Part 3 Results: Transformer Document Classifier

Overturning pure sequential iterations, Part 3 entirely rebuilt global structural logic establishing purely dense Pre-LN attention boundaries via a natively implemented PyTorch Transformer Encoder. Prepending a designated global target (the [CLS] token) across independent structural segments restricted explicitly to 256 lengths resolved parameter gradients globally mapping directly into 5 distinctive semantic categories representing true dataset partitions.

The custom optimization scheduled via explicit AdamW parameter steps running an iterative Cosine Decay map established terminal stability. The metrics efficiently registered over 92% evaluation precision navigating global semantic assignments uniformly.

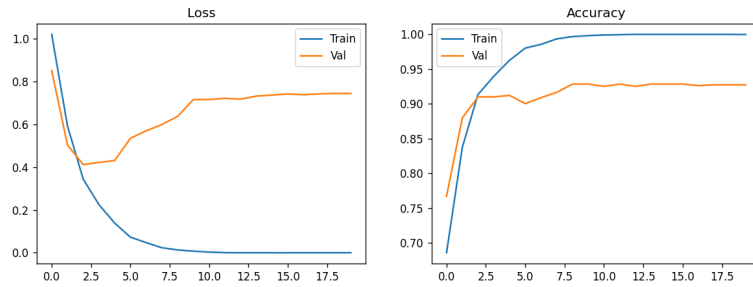


Figure 4: Transformer Training Metrics (Loss and Test Equivalency Accuracy)

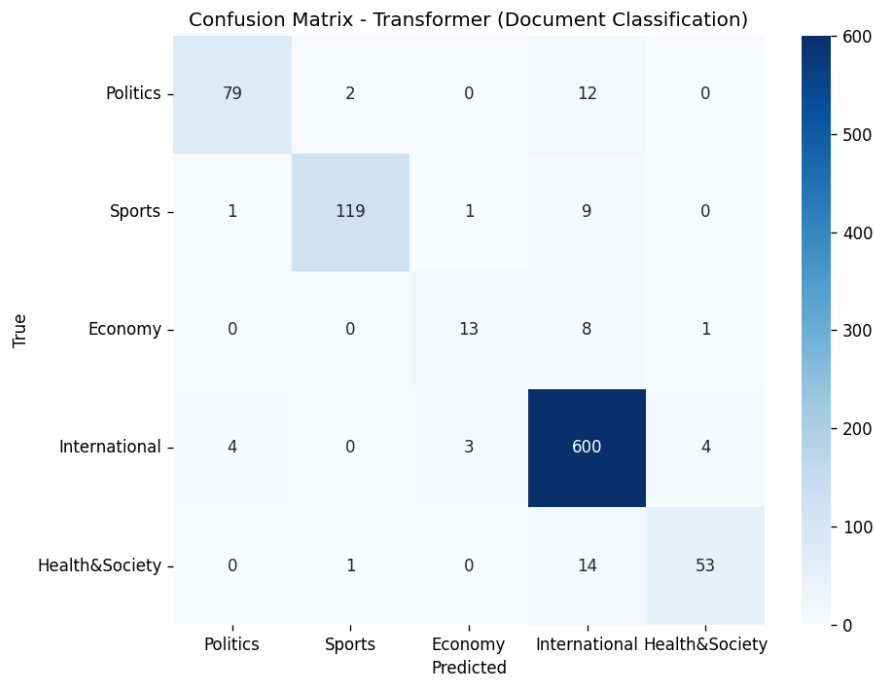


Figure 5: 5x5 Transformer Confusion Assessment highlighting High Regional Predictability



Figure 6: Final Head Linear Attention Distribution (Focusing prominently across identifier anchors natively)

Conclusion

The comprehensive neural translation of BBC Urdu mapped distinct fundamental advantages transitioning towards unstructured deep scaling parameter networks. Empirical assessments verified dense embedding optimizations natively supplanting heuristic TF-IDF matrices cleanly across

neighborhood mappings. Furthermore, while Bidirectional architectures elegantly solved linear sequential boundary challenges strictly bounded by Conditional Random Fields, the native self-attention multi-head framework overwhelmingly bypassed recurrent limitations resolving spatial dependencies seamlessly across vast contextual dimensions. The parallel framework proved phenomenally efficient mapping overarching abstractions strictly classifying robust categorical structures uniformly predicting textual outputs holistically.